INCH-POUND
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SUPERSEDING
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DETAIL SPECIFICATION

CORD, FIBROUS, POLYESTER, SOLID BRAID

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a solid braided polyester coreless cord.

1.2 Classification. The braided cord is of one type, and in the diameters specified in 3.5 (see 6.2)

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

DEPARTMENT OF DEFENSE STANDARDS

FEDERAL

FED-STD-191 - Textile Test Methods

COMMERCIAL ITEM DESCRIPTION

A-A-59622 - Tags, Shipping and stock

(Copies of these documents are available online at https://assist.dla.mil).

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue Philadelphia, PA 19111-5096 or e-mail to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes.

(Copies of this documents are available online at https://www.asq.org or the American Society for Quality, 600 North Plankinton Avenue, Milwaukee, WI 53203.)

SAE INTERNATIONAL (SAE)

SAE AMS-STD-595 Colors Used in Government Procurement

(Copies of these documents are available from www.sae.org or SAE International Customer Service, 400 Commonwealth Drive, Warrendale, PA 15096.)

2.4 Order of Precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.2.

3.2 Government and contractor purchases. The requirements specified in 3.8 and 3.9 apply only to cord purchased directly by the Government. All other requirements apply both to cord purchased by a contractor as a component for an end item and to cord purchased directly by the Government.

3.3 Standard sample. The cord shall match the standard sample for shade and shall be equal to or better than the standard sample with respect to all characteristics for which the standard sample is referenced (see 6. 4).

3.4 Material. The yarn used in the fabrication of the cord shall be a multifilament, plied or single polyester yarn. When dyed cord specified, the shade shall be obtained by yarn dyeing or piece dyeing and subsequently the material shall be processed to stabilize the yarn or cord (see 6.5). It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.5 Construction and physical requirements. The cord shall be a solid braid construction without a core. The fabricated cord shall have a compact structure of locked stitches forming a round, even, solid cord and shall be processed to meet the requirements specified in table I when tested as specified in 4.3.3. The character of the cord shall be equal to the standard sample for roundness and firmness (see 6.4)

3.6 Color. Unless otherwise specified (see 6.2), the color of the cord shall be natural. When colored cord is specified, the cord shall be dyed Green 383 Camo, to approximate color chip 34094 of SAE AMS-STD-595 (see 6.6).

3.6.1 Colorfastness. The dyed cord shall show fastness to weathering equal to or better than the standard sample or equal to or better than a rating of "good" when tested as specified in 4.3.3.

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TABLE I. Physical requirements.

Diameter In inches (+.0156)	Number of strands (minimum)	Breaking strength, pounds (minimum)	Length per pound, feet (minimum)	Elongation percent (maximum)
.0938	8	205	360	25
.125	9	310	220	25
.1563	12	400	140	25
.1875	12	540	99	25
.2188	12	725	71	25
.250	12	945	53	25
.3125	12	1575	34	25
.375	12	1925	25	25
.500	12	2950	14	25

3.6.2 Matching. The color of the dyed cord shall match the standard sample when viewed under filtered tungsten lamps that approximate artificial daylight and that have a correlated color temperature of 7500 ± 200K, with illumination of 100 ± 20 foot candles, and shall be a good match to the standard sample under incandescent lamplight at 2300 ± 200K.

3.7 Non-fibrous material (chloroform-soluble material). No extraneous material shall be added for the purpose of weighting the cord. The chloroform soluble material of the cord shall not exceed 4.0 percent when tested as specified in 4.3.3.

3.8 Put-up. Unless otherwise specified (see 6.2), the cord shall be furnished without knots or splices on spools (reels) in the minimum lengths specified in table II. A plus tolerance of 10.0 percent shall be allowed on the length of spool (reel) specified. Broken lengths will be permitted, but a spool (reel) shall contain not more than three pieces, and no piece shall be less than 100 feet in length, or no more than two pieces with no applicable minimum length. The cord shall be wound on the spools (reels) so that each turn and layer is free from entanglement. The ends of the cord shall be heat sealed to prevent fraying.

3.9 Identification. Each spool (reel) of cord shall have a ticket (identification tag) or label attached to it for identification purposes. The ticket shall be in accordance with A-A-59622. When labels are used, the label shall be attached in such a manner as to remain in place and be clearly legible until all cord has been removed. All ticket or label entries shall be legibly printed, stamped, or typed with water insoluble ink. The ticket or label shall contain the following information:

Stock Number
Nomenclature
Size (diameter)
Specification number
Length
Contract number and date
Contractor's name

3.10 Workmanship. The end item shall conform to the quality of product established by this specification.

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TABLE II. Put-up spool reel length.

Diameter In inches	Length of spool (reel) feet (minimum)
.0938	3000
.125	3000
.1563	3000
.1875	3000
.1875	100 1/
.2188	2000
.250	2000
.3125	1000
.375	1000
.500	1000

1/ Only one piece permitted.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).
b. Conformance inspection (see 4.3).

4.2 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.3.2.1 and 4.3.2.2 and tested for the characteristics specified in 4.3.3.

4.3 Conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with ASQ-Z1.4.

4.3.1 Component and material inspections. Components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.1.1 Component and material certification. A certificate of compliance may be acceptable as evidence that the material conforms to the requirements specified in 3.4.

4.3.2 End item examination.

4.3.2.1 End item visual examination. The cord shall be examined for the defects listed in table III. All defects shall be counted regardless of their proximity to each other. The lot size shall be expressed in units of spools or reels of cord. The sample unit shall be one spool or reel of cord. Ten percent of the total footage, but not less than 100 feet, shall be examined on each spool or reel of cord in the sample. The inspection level shall be II.

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Table III. End item visual defects.

Examine	Defect	Classification Major	Minor
Appearance and workmanship	Any cut	101	
	Abraded surface, chafed or damaged	102	
	Kinks or unevenly braided resulting in open places, breaks in continuity of braids, or soft spots	103	
	Loosely braided	104	
	Broken, loose, or projecting strand or end	105	
	Not equal to standard sample for roundness and firmness	106	
Color	Other than specified	107	
	Not uniform		201
Cleanness	Spot or stain clearly visible 1/		202
Identification ticket or label	Omitted, incorrect, illegible, insecurely attached		203
	Ticket or label not as specified		204

1/ At normal inspection distance (approximately 3 feet).

4.3.2.2 Length and winding examination. The end item shall be examined for the defects listed below. The lot size shall be expressed in units spools or reels of cord. The sample unit shall be one spool or reel of cord. For lots consisting of 500 or fewer units, the sample size shall be 10. Defects shall be as listed below. The inspections level shall be S-3.

Defects

Length less than specified or more than 10 percent in excess of length specified.

Length of cord on the spool (reel) is less than marked on the ticket or label.

Improperly or not firmly wound resulting in kinking, knotting, entangling, or slippage during unwinding, or otherwise affecting free unhampered unwinding of cord.

Put-up not as specified.

Any end not heat sealed.

Any spool (reel) found to contain more than three pieces.

Any piece less than 100 feet in length (except when there are only two pieces per spool).

Any knot or splice.

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4.3.3 End item testing. The methods of testing specified in FED-STD-191, wherever applicable, and as listed in table IV shall be followed. When the data in the "Number of determinations" and "Results reported as" columns are not specified in table IV, they shall be as required by the referenced test method. The physical and chemical values specified in section 3 apply to the average of the determinations made on a sample unit for test purposes as specified in the applicable test methods. The lot size shall be expressed in units of spools or reels of cord, as applicable. The sample unit shall be one spool or reel of cord. The sample size shall be in accordance with table V. All test reports shall contain the individual values utilized in expressing the final results. The lot shall be unacceptable if one or more units fail to meet any requirement specified. Tests to determine compliance with specification requirements, including quantity of delivery, may be made under prevailing atmospheric conditions. In cases of dispute, tests shall be made upon material which has reached equilibrium under standard conditions as defined in FED-STD-191.

TABLE IV. End Item tests.

Characteristics	Requirement paragraph	Test method	No. of deter. per sample unit	Results reported as
Construction (solid braid without core)	3.5	Visual	1	Pass or fail
Diameter	3.5	6002	-	-
Number of strands	3.5	Visual	1	Pass or fail
Breaking strength and elongation	3.5	6016	-	-
Length per pound	3.5	6004	-	-
Colorfastness to weathering	3.6.1	5671.1 1/	-	Pass or fail
Nonfibrous material (chloroform-soluble material)	3.7	2611	-	-

1/ The time of exposure shall be 40 standard fading hours.

TABLE V. Sample size.

Lot size (spools or reels)	Sample size (sample units)
800 or less	2
801 up to and including 22,000	3
22,000 and over	5

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5. PACKAGING

5.1. Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The cord is intended for use in tentage, equipage, and other such items, and also for orthopedic and neurosurgical procedures.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Diameter of cord required (see 1.2 and 3.5).
c. When first article is required (see 3.1, 4.2 and 6.3).
d. When color Camouflage Green 383 is required (see 3.6).
e. Put-up, if other than specified (see 3.8).
f. Packaging requirements (see 5.1).

6.3 First article. When a first article is required, it should be inspected and approved under the appropriate provisions of FAR 52.209-4. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Sample. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Twenty-five percent elongation. The maximum elongation requirement of 25 percent has been successfully met by:

a. The use of yarn dyed polyester stabilized prior to braiding.
b. Heat stretching of piece dyed cord.

6.6 Color. Olive green and olive drab colors have been replaced by Camouflage Green 383. Any item which previously required any olive green or olive drab color will now require Camouflage Green 383.

6.7 Subject term (key word) listing.

Cordage
Equipage
Neurosurgical
Orthopedic
Tent

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Army - GL
Navy - MC
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Review activities:
Army - CR4, MD, MI

Preparing Activity:
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